

Project Design Phase-II

Technology Stack (Architecture & Stack)

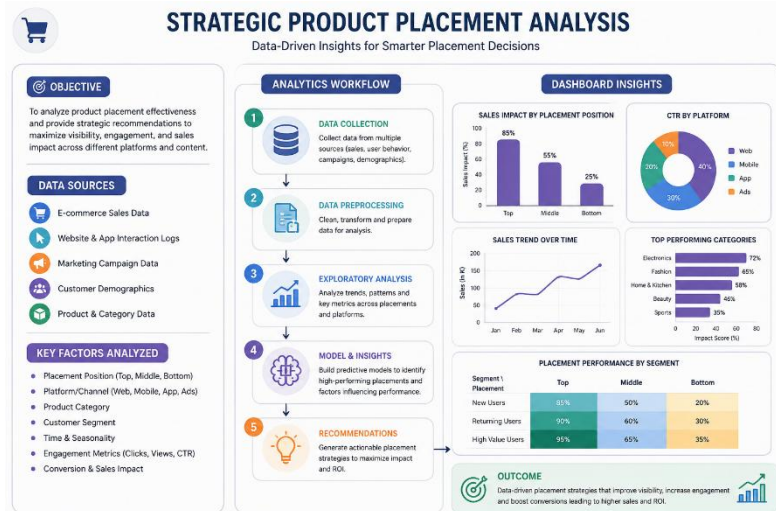
Date	
Team ID	
Project Name	
Maximum Marks	

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

Component	Technology / Tool Used	Purpose
Frontend	HTML5	Creates the web page structure for the dashboard.
Styling	CSS3	Designs the user interface and improves appearance.
Backend	Python	Handles application logic and serves web pages.
Web Framework	Flask	Connects the frontend with Tableau dashboard and manages routing.
Data Source	CSV Dataset	Stores product placement and sales data for analysis.
Data Visualization	Tableau Public	Creates interactive charts, dashboards, and stories.
Dashboard Integration	Tableau Embed API	Embeds the Tableau dashboard into the web application.
Development Environment	Visual Studio Code	Used for coding, editing, and debugging the project.
Version Control	Git	Tracks project changes and manages source code.
Repository Hosting	GitHub	Stores the project online and enables collaboration.
Browser	Google Chrome / Microsoft Edge	Displays the Flask web application and Tableau dashboard.
Operating System	Windows 10/11	Development and execution environment.
Data Processing	Tableau Calculated Fields	Performs calculations such as Average Sales, Profit, and Sales Metrics.
Dashboard Features	Tableau Filters	Allows users to filter data by Product Category, Season, Product Position, Promotion, etc.

Component	Technology / Tool Used	Purpose
Documentation	Microsoft Word	Used to prepare project documentation and reports.

Table-2: Application Characteristics:

Characteristic	Description
Application Name	Strategic Product Placement Analysis
Application Type	Data Analytics & Business Intelligence Dashboard
Domain	Retail Analytics
Purpose	Analyze product placement, pricing, promotions, and customer behavior to improve sales performance.
Target Users	Retail Managers, Store Managers, Marketing Team, Business Analysts
Platform	Web-based Dashboard
Frontend	HTML5, CSS3
Backend	Python Flask
Visualization Tool	Tableau Public
Programming Language	Python
Database/File Source	CSV Dataset
Operating System	Windows 10/11

Characteristic	Description
Browser Support	Google Chrome, Microsoft Edge, Mozilla Firefox
Data Processing	Data Cleaning, Aggregation, Filtering, Calculated Fields
Reports	Interactive Dashboard, Story Points, Charts, Filters
Security	Read-only dashboard access, secure local project files
Performance	Fast dashboard loading with interactive visualizations
Scalability	Can handle additional datasets and more product categories
Availability	Accessible through Tableau Public and Flask Web Interface
Usability	Simple and user-friendly dashboard with interactive filters
Maintainability	Easy to update datasets, dashboards, and visualizations
Portability	Can be deployed on local machine or cloud server
Project Outcome	Helps retailers identify the best product placement strategies and improve sales decisions.

References:

<https://code.visualstudio.com/docs>

<https://help.tableau.com/current/pro/desktop/en-us/default.htm>

<https://docs.github.com/en>

<https://docs.python.org/3/>

